GAL. - Temă seminar 4

1\. a) $S_3 = \{ v_1 = (1,1,0),\ v_2 = (1,0,1),\ v_3 = (0,1,1) \} \subset \mathbb{R}^3/\mathbb{R}$

$S_3 \subset \mathbb{R}^3/\mathbb{R}$ sis de gen $\Leftrightarrow$ $(\forall)\ v \in \mathbb{R}^3$, $(\exists)\ \alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$

a.î. $v = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3$

Fie $v = (x, y, z) \in \mathbb{R}^3$
vector arbitrar

$v = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 \Leftrightarrow$

$(x, y, z) = (\alpha_1 + \alpha_2,\ \alpha_1 + \alpha_3,\ \alpha_2 + \alpha_3)$

$$\begin{cases} \alpha_1 + \alpha_2 = x \\ \alpha_1 + \alpha_3 = y \\ \alpha_2 + \alpha_3 = z \end{cases} \Rightarrow 2(\alpha_1 + \alpha_2 + \alpha_3) = x + y + z \Rightarrow \begin{cases} \cancel{\alpha_1 = x + y} \\ \cancel{\alpha_2 = x + z} \\ \cancel{\alpha_3 = y + z} \end{cases}$$

$$\Rightarrow \alpha_1 + \alpha_2 + \alpha_3 = \frac{x + y + z}{2} \Rightarrow \begin{cases} \alpha_1 = \dfrac{x + y - z}{2} \\ \alpha_2 = \dfrac{x - y + z}{2} \\ \alpha_3 = \dfrac{-x + y + z}{2} \end{cases}$$

deci $(\exists)$
$$\begin{cases} \alpha_1 = \dfrac{x + y - z}{2} \\ \alpha_2 = \dfrac{x - y + z}{2} \\ \alpha_3 = \dfrac{-x + y + z}{2} \end{cases}$$
a.î. $(\forall)\ v = (x, y, z) \in \mathbb{R}^3 \Rightarrow$
$v = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3$

$\Rightarrow S_3 = \{ v_1, v_2, v_3 \} \subset \mathbb{R}^2/\mathbb{R}$ sist de gen pt $\mathbb{R}^2/\mathbb{R}$

b) $S'_4 = \{v_1 = 1,\ v_2 = X-1,\ v_3 = (X-1)^2\} \subset \mathbb{R}_2[X]$

$S'_4 \subset \mathbb{R}_2[X]$ sis de gen $\Leftrightarrow$ $(\forall)\ p \in \mathbb{R}_2[X]$, $(\exists)\ \alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$

$p = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3$

Fie $p = aX^2 + bX + c \in \mathbb{R}_2[X]$
polinom arbitrar

$p = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 \Leftrightarrow$

$aX^2 + bX + c = \alpha_3 X^2 + (\alpha_2 - 2\alpha_3) X + \alpha_1 - \alpha_2 + \alpha_3$

$\begin{cases} a = \alpha_3 \\ b = \alpha_2 - 2\alpha_3 \\ c = \alpha_1 - \alpha_2 + \alpha_3 \end{cases} \Rightarrow \begin{array}{l} \alpha_3 = a \\ \alpha_2 = b + 2a \\ \alpha_1 = c + b + 2a - a = a + b + c \end{array}$

deci $(\exists)$ $\begin{cases} \alpha_3 = a \\ \alpha_2 = 2a + b \\ \alpha_1 = a + b + c \end{cases}$ a.î. $(\forall)\ p = aX^2 + bX + c \in \mathbb{R}_2[X]$

$p = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 \Rightarrow$

$\Rightarrow S'_4 = \{v_1, v_2, v_3\} \subset \mathbb{R}_2[X]$ sis de gen (pt $\mathbb{R}_2[X]$)